ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2025

Assignment 6 - Due date 02/27/25

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Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., "LuanaLima_TSA_A06_Sp25.Rmd"). Then change "Student Name" on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages needed for this assignment: "ggplot2", "forecast", "tseries" and "sarima". Install these packages, if you haven't done yet. Do not forget to load them before running your script, since they are NOT default packages.

```
#Load/install required package here
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
library(ggplot2)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(Kendall)
library(tseries)
library(outliers)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr   1.1.4     v stringr 1.5.1
## v forcats 1.0.0     v tibble  3.2.1
## v purrr   1.0.2     v tidyr   1.3.1
## v readr   2.1.5

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
#install.packages("sarima")
library(sarima)
```

```
## Loading required package: stats4
##
## Attaching package: 'sarima'
##
## The following object is masked from 'package:stats':
##
##     spectrum
```

This assignment has general questions about ARIMA Models.

Q1

Describe the important characteristics of the sample autocorrelation function (ACF) plot and the partial sample autocorrelation function (PACF) plot for the following models:

- AR(2)

  Answer: The ACF will likely show a slow decay. The PACF will have a cut off at lag 2.

- MA(1)

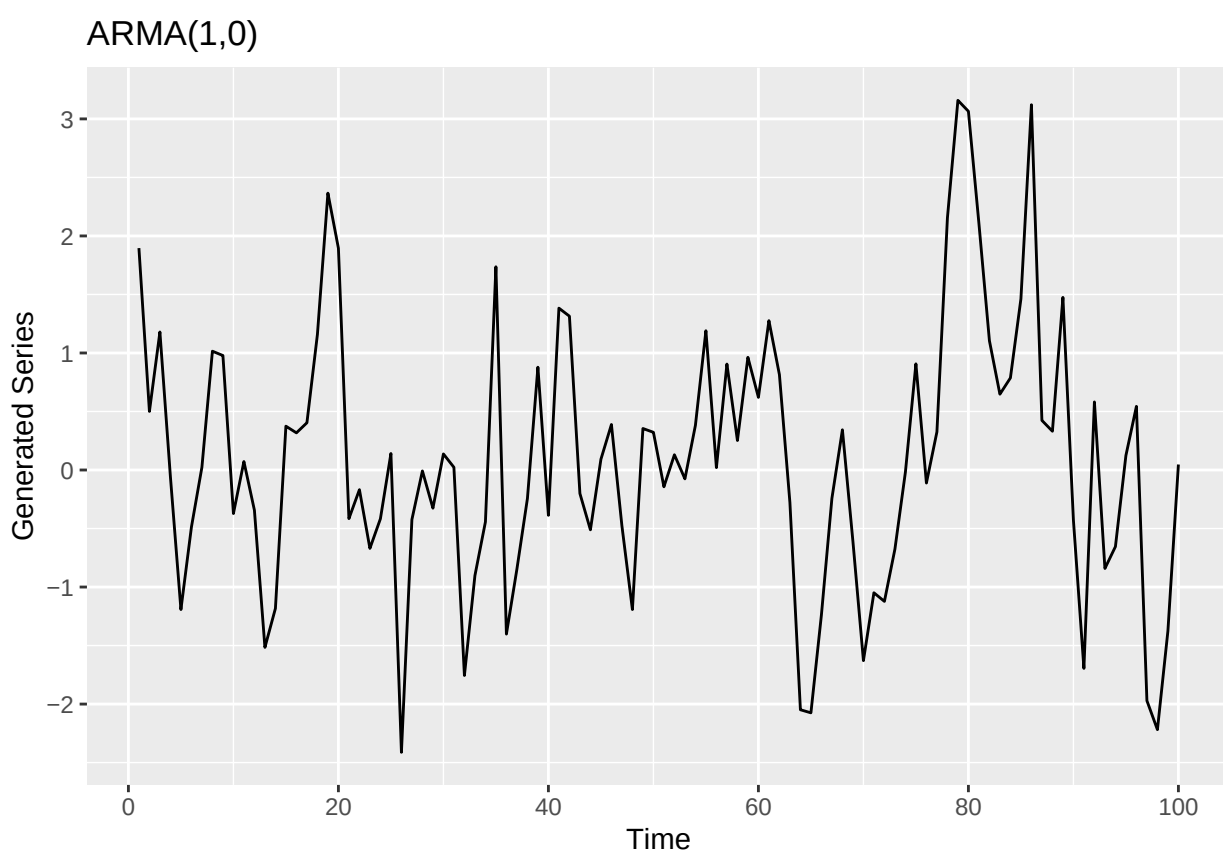
  Answer: The ACF will have a significant value at lag 1. After that the values may oscillate but will not be significant. The PACF may show a slow decay.

Q2

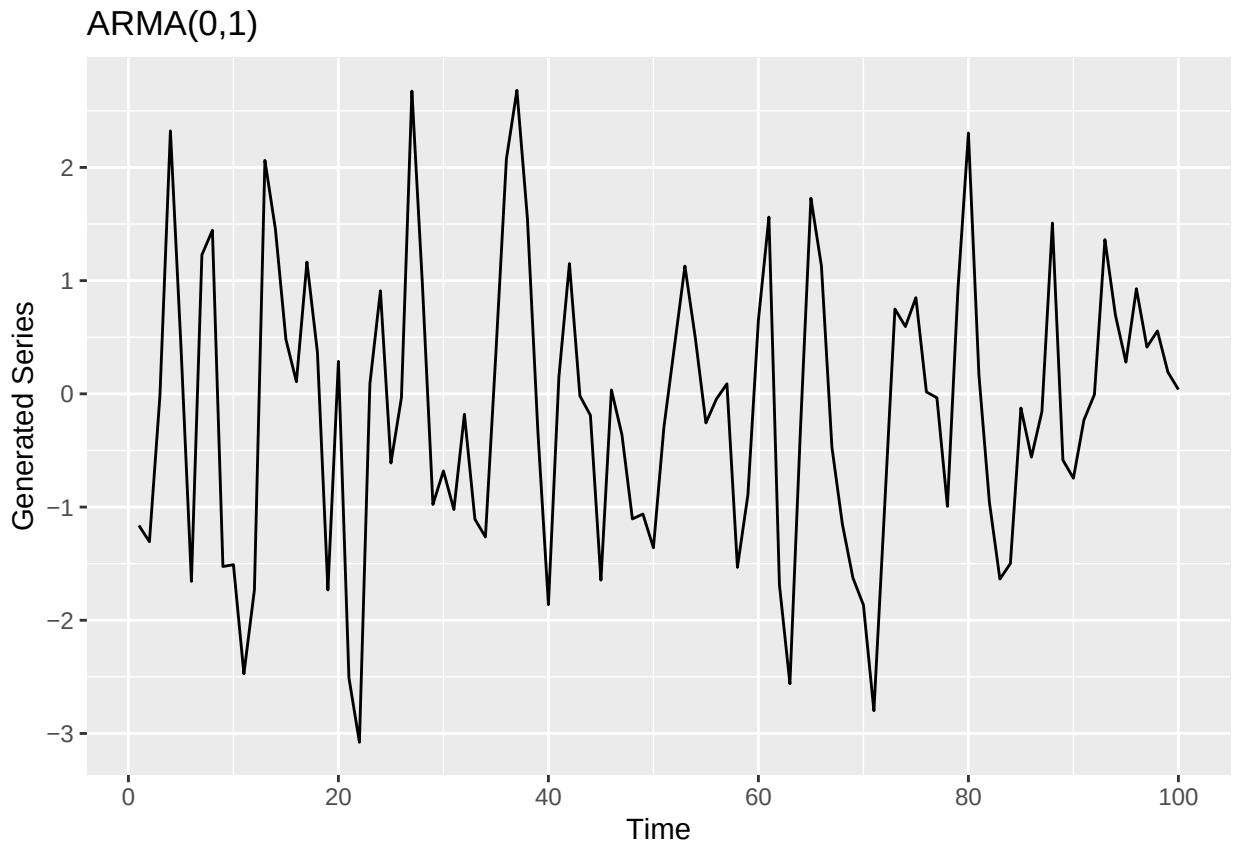
Recall that the non-seasonal ARIMA is described by three parameters ARIMA(p, d, q) where p is the order of the autoregressive component, d is the number of times the series need to be differenced to obtain stationarity and q is the order of the moving average component. If we don't need to difference the series, we don't need to specify the "I" part and we can use the short version, i.e., the ARMA(p, q).

(a) Consider three models: ARMA(1,0), ARMA(0,1) and ARMA(1,1) with parameters $\phi = 0.6$ and $\theta = 0.9$. The ϕ refers to the AR coefficient and the θ refers to the MA coefficient. Use the `arima.sim()` function in R to generate $n = 100$ observations from each of these three models. Then, using `autoplot()` plot the generated series in three separate graphs.

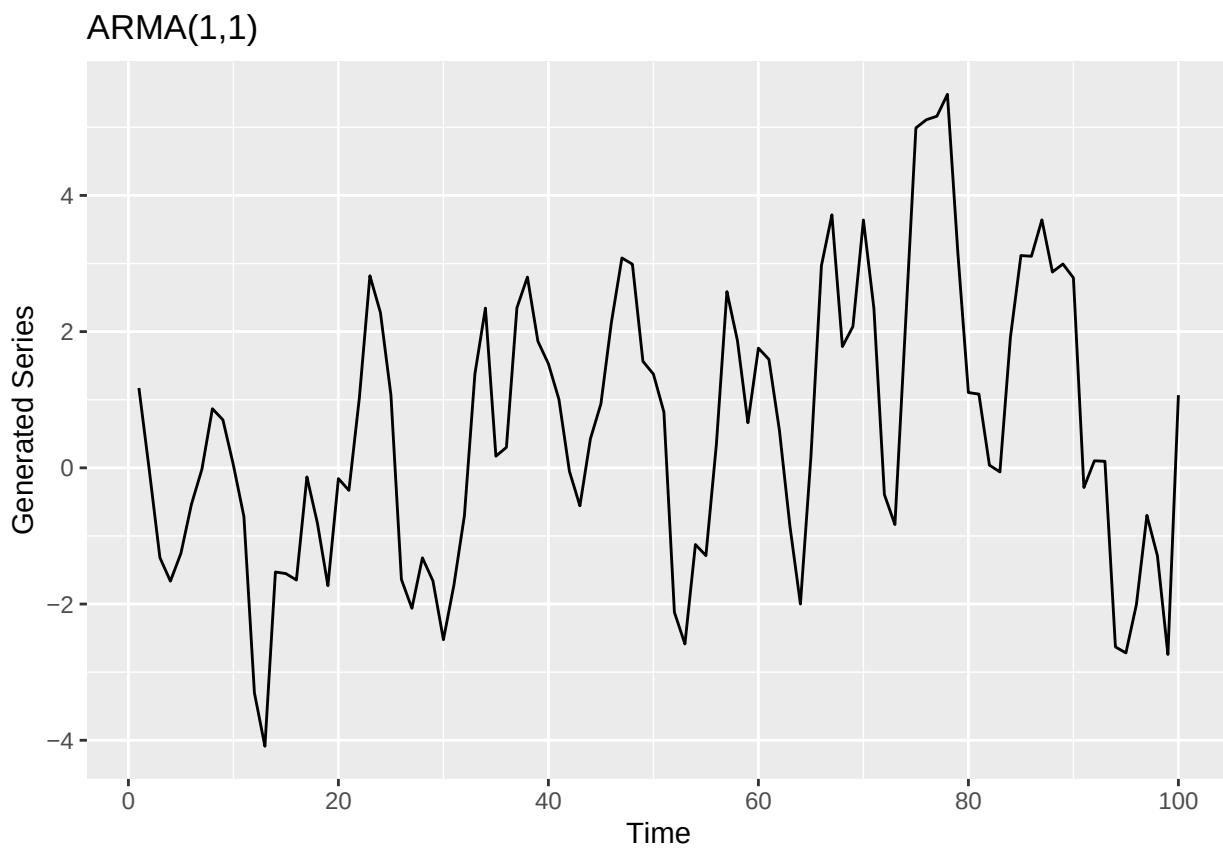
```
ARMA10 <- arima.sim(list(order=c(1,0,0),ar=c(0.6)),n=100)
ARMA01 <- arima.sim(list(order=c(0,0,1),ma=c(0.9)),n=100)
ARMA11 <- arima.sim(list(order=c(1,0,1),ar=c(0.6),ma=c(0.9)),n=100)

autoplot(ARMA10)+
  ggtitle("ARMA(1,0)")+
  ylab("Generated Series")
```




```
autoplot(ARMA01)+
  ggtitle("ARMA(0,1)")+
  ylab("Generated Series")
```



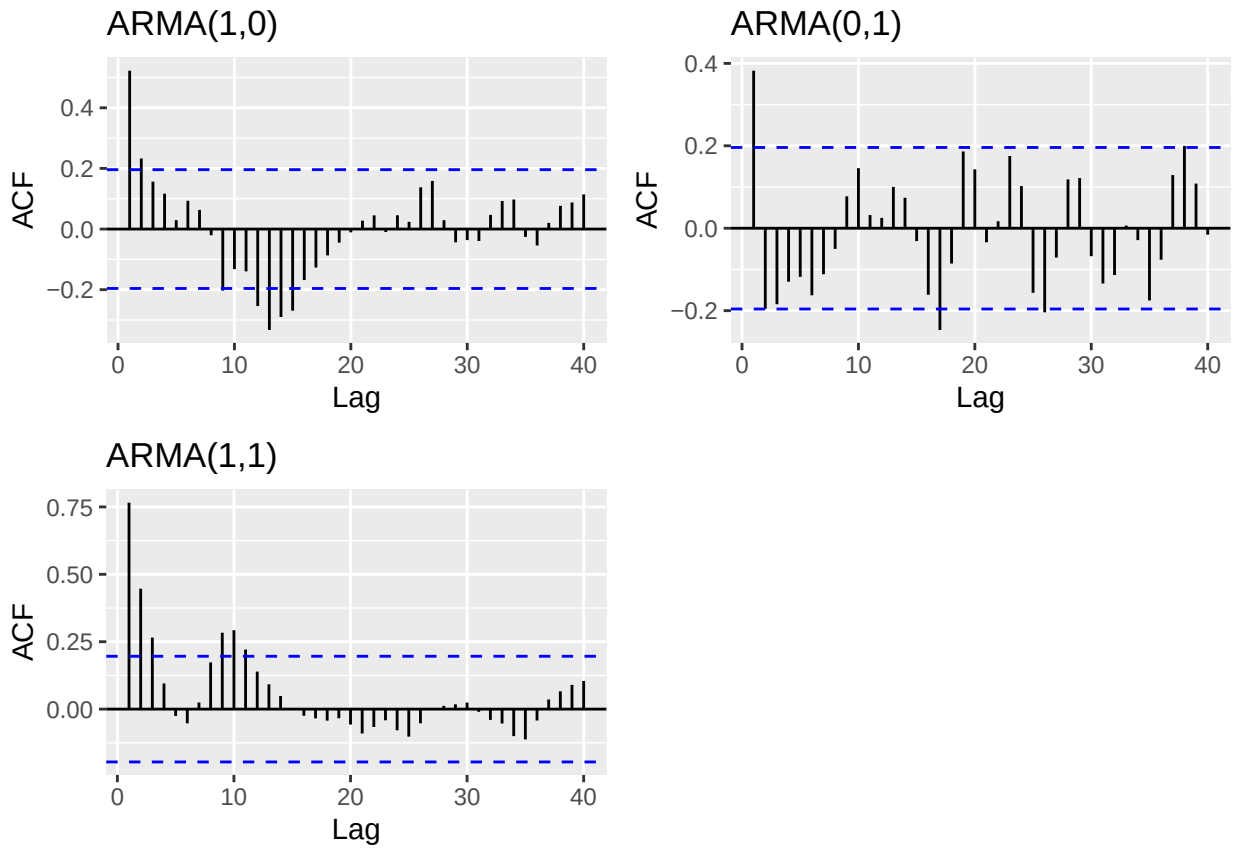

```
autoplot(ARMA11)+
  ggtitle("ARMA(1,1)")+
  ylab("Generated Series")
```



(b) Plot the sample ACF for each of these models in one window to facilitate comparison (Hint: use `cowplot::plot_grid()`).

```
plot_grid(
  autoplot(Acf(ARMA10, lag.max=40, plot = FALSE),main = "ARMA(1,0)"),
  autoplot(Acf(ARMA01, lag.max=40, plot = FALSE), main = "ARMA(0,1)"),
  autoplot(Acf(ARMA11, lag.max=40, plot = FALSE),main = "ARMA(1,1)")
)
```

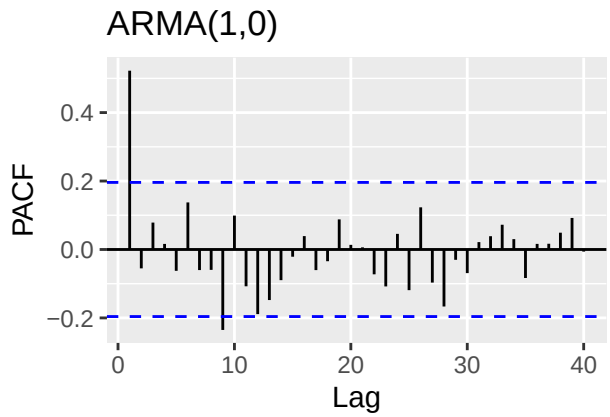
```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown parameters: `main`
## Ignoring unknown parameters: `main`
## Ignoring unknown parameters: `main`
```

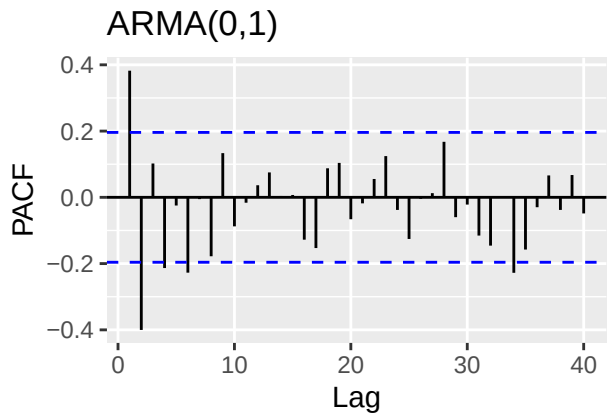


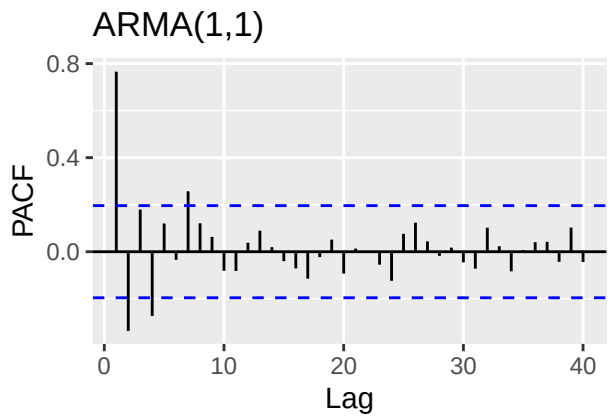
(c) Plot the sample PACF for each of these models in one window to facilitate comparison.

```
plot_grid(
  autoplot(Pacf(ARMA10, lag.max=40, plot = FALSE),main = "ARMA(1,0)"),
  autoplot(Pacf(ARMA01, lag.max=40, plot = FALSE), main = "ARMA(0,1)"),
  autoplot(Pacf(ARMA11, lag.max=40, plot = FALSE),main = "ARMA(1,1)")
)
```

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown parameters: `main`
## Ignoring unknown parameters: `main`
## Ignoring unknown parameters: `main`
```





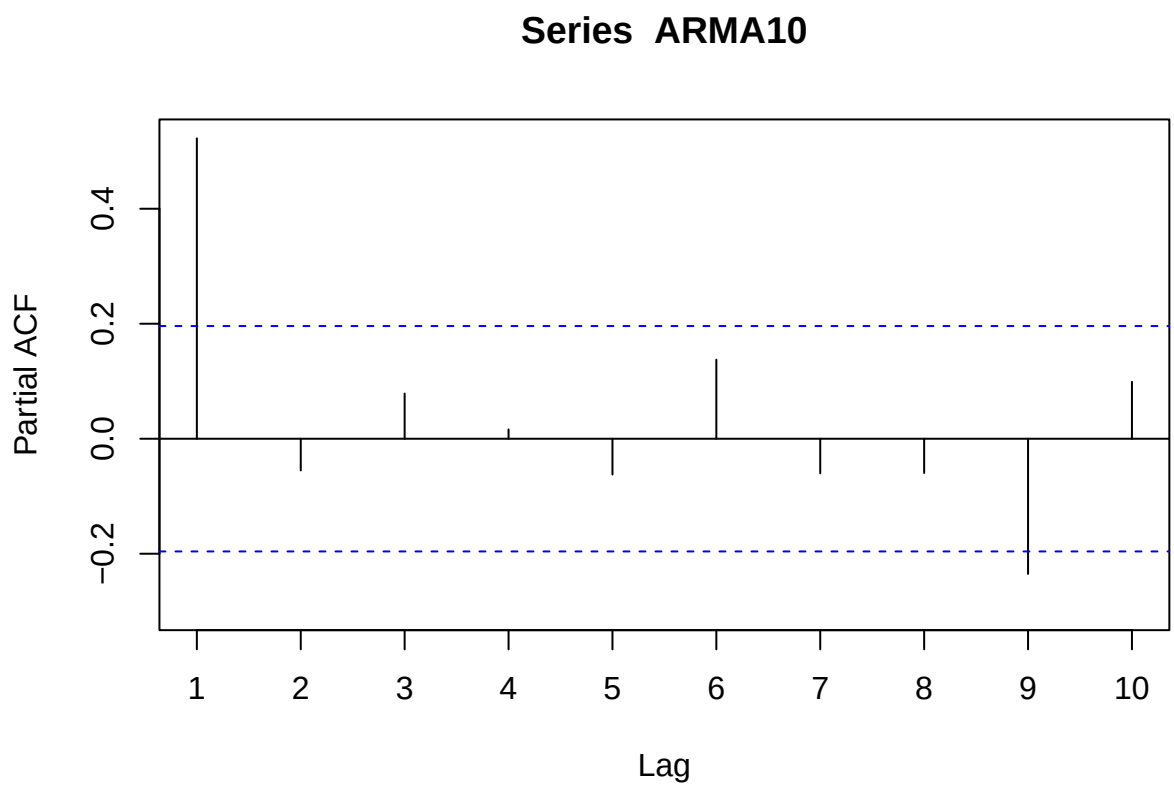


(d) Look at the ACFs and PACFs. Imagine you had these plots for a data set and you were asked to identify the model, i.e., is it AR, MA or ARMA and the order of each component. Would you be able identify them correctly? Explain your answer.

Answer:The models with either AR or MA components are somewhat identifiable though could cause some confusion. For ARMA(1,0), the PACF shows a significant spike at lag 1 and then the rest of the values are within the blue lines. For purely AR models, the PACF tells us what order the model is based on the cut off value. Here, the cutoff and the AR order are 1. The ACF for this model does not have a very obvious slow decay, but there is still some decay rather than a clear cut off so I would likely have identified this model as AR(1) For ARMA(0,1), the ACF seems to have a cut off at lag 1 but there are still some significant peaks after. The PACF does not have a clear cut off either and seems to have a slow decay. Because neither has a clear cut off I might have identified this as an ARMA model. The ARMA(1,1) ACF could be interpreted as having 2 significant peaks and the PACF could be interpreted as having 2 significant peaks. Purely AR or MA models don't have significant peaks for both ACF and PACF so I would have identified this as an ARMA model, although, as with all ARMA models, I wouldn't have been able to identify the order from the ACF and PACF.

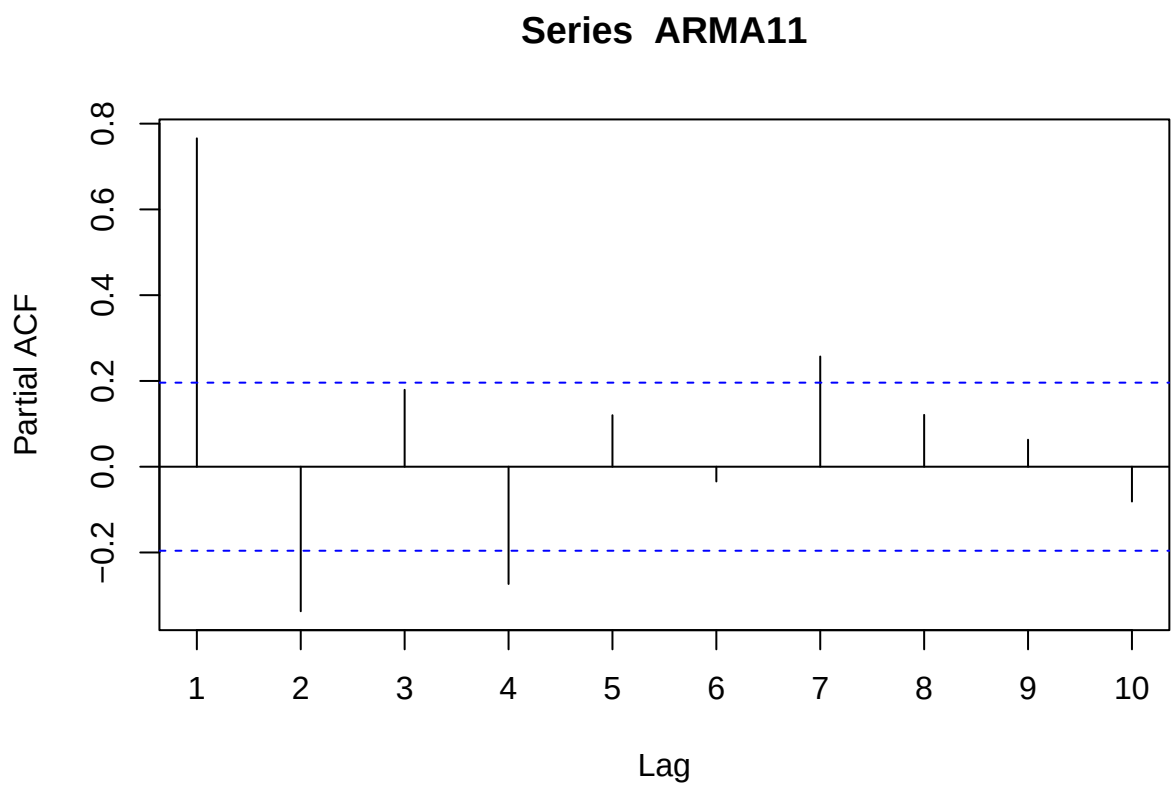
(e) Compare the PACF values R computed with the values you provided for the lag 1 correlation coefficient, i.e., does $\phi = 0.6$ match what you see on PACF for ARMA(1,0), and ARMA(1,1)? Should they match?

```
print(Pacf(ARMA10, lag.max = 10))
```




```
##
## Partial autocorrelations of series 'ARMA10', by lag
##
##      1      2      3      4      5      6      7      8      9     10
##  0.522 -0.055  0.079  0.016 -0.062  0.137 -0.060 -0.060 -0.235  0.099
```

```r
print(Pacf(ARMA11, lag.max = 10))
```

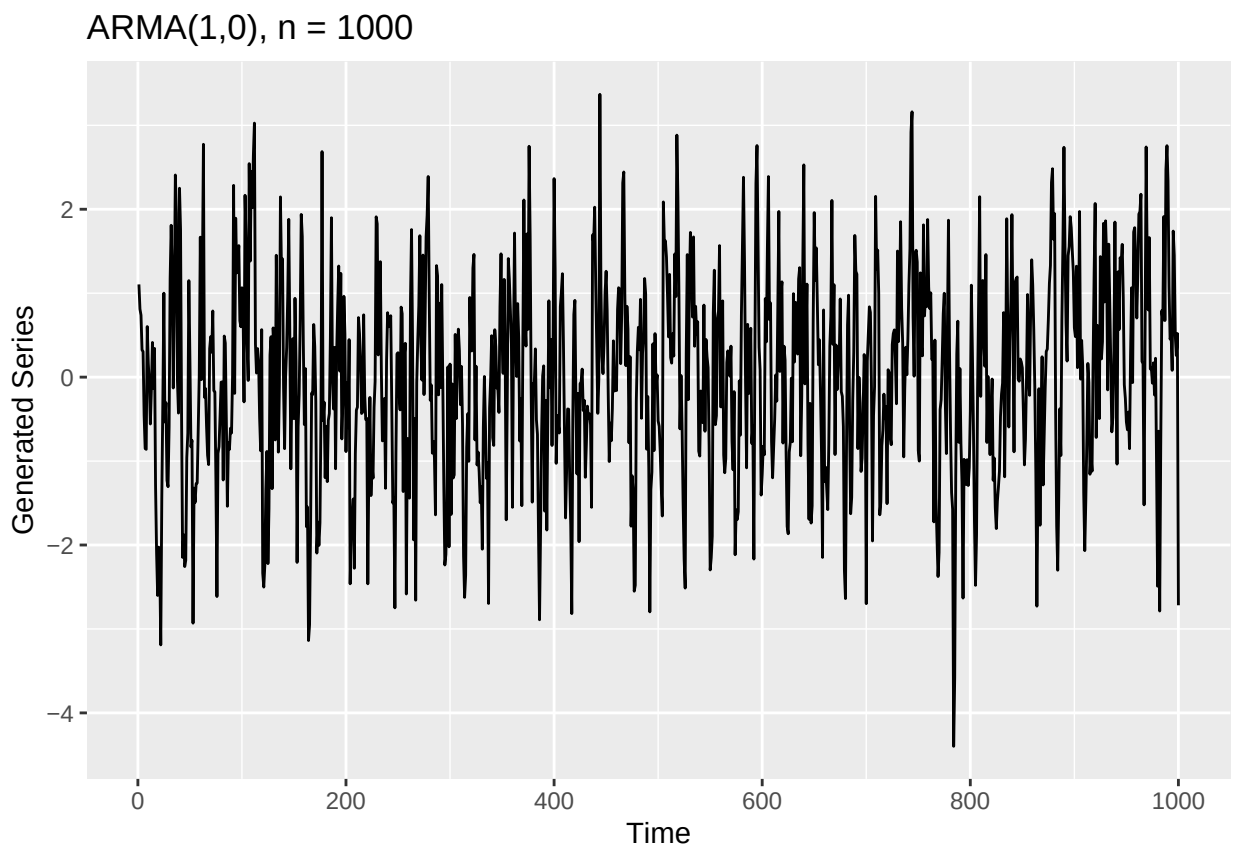



```
##
## Partial autocorrelations of series 'ARMA11', by lag
##
##      1      2      3      4      5      6      7      8      9     10
##  0.766 -0.337  0.179 -0.273  0.120 -0.034  0.257  0.121  0.063 -0.081
```

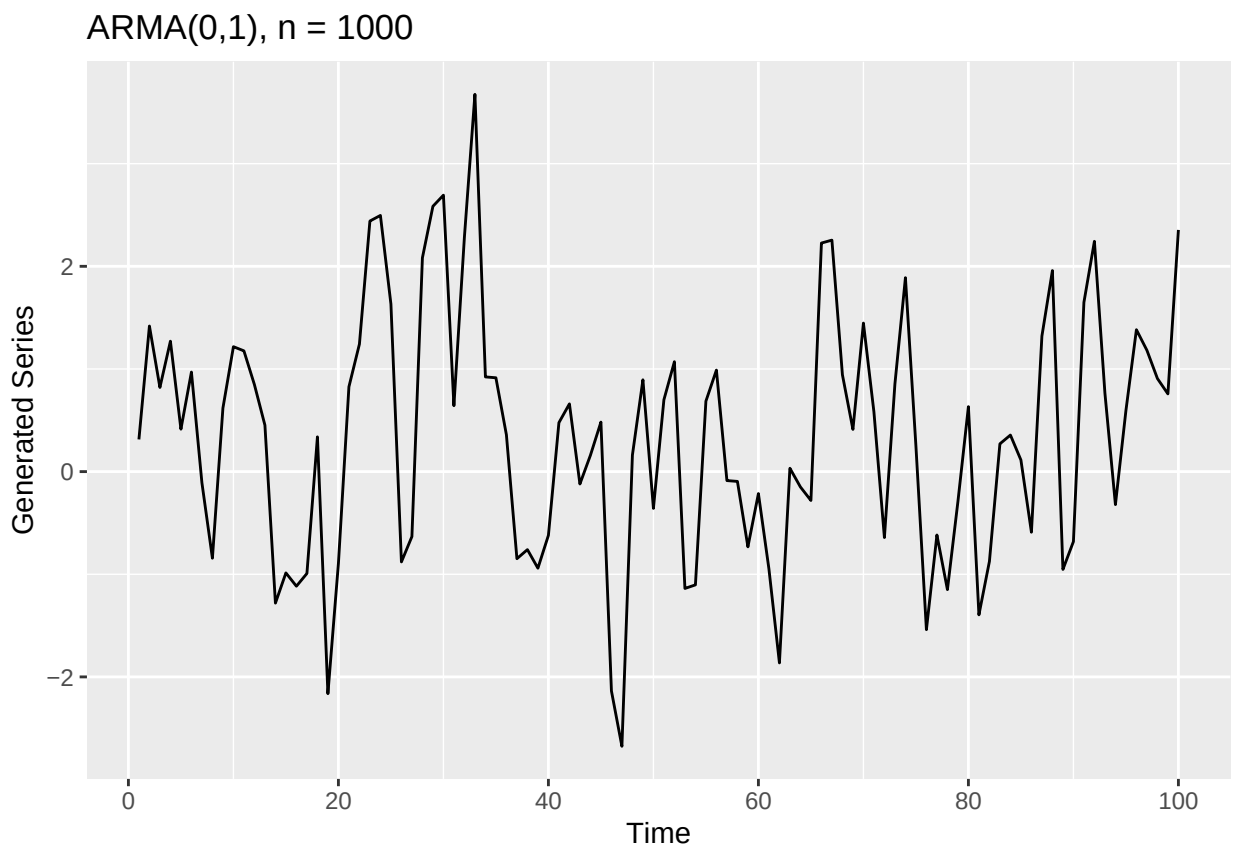
Answer: The PACF values for ARMA (1,0) and ARMA(1,1) do not match. This is expected because while the ARMA(1,1) model does incorporate the AR component with the 0.6 coefficient and error term, it also includes the MA component and it's coefficient (0.9).

(f) Increase number of observations to $n = 1000$ and repeat parts (b)-(e).

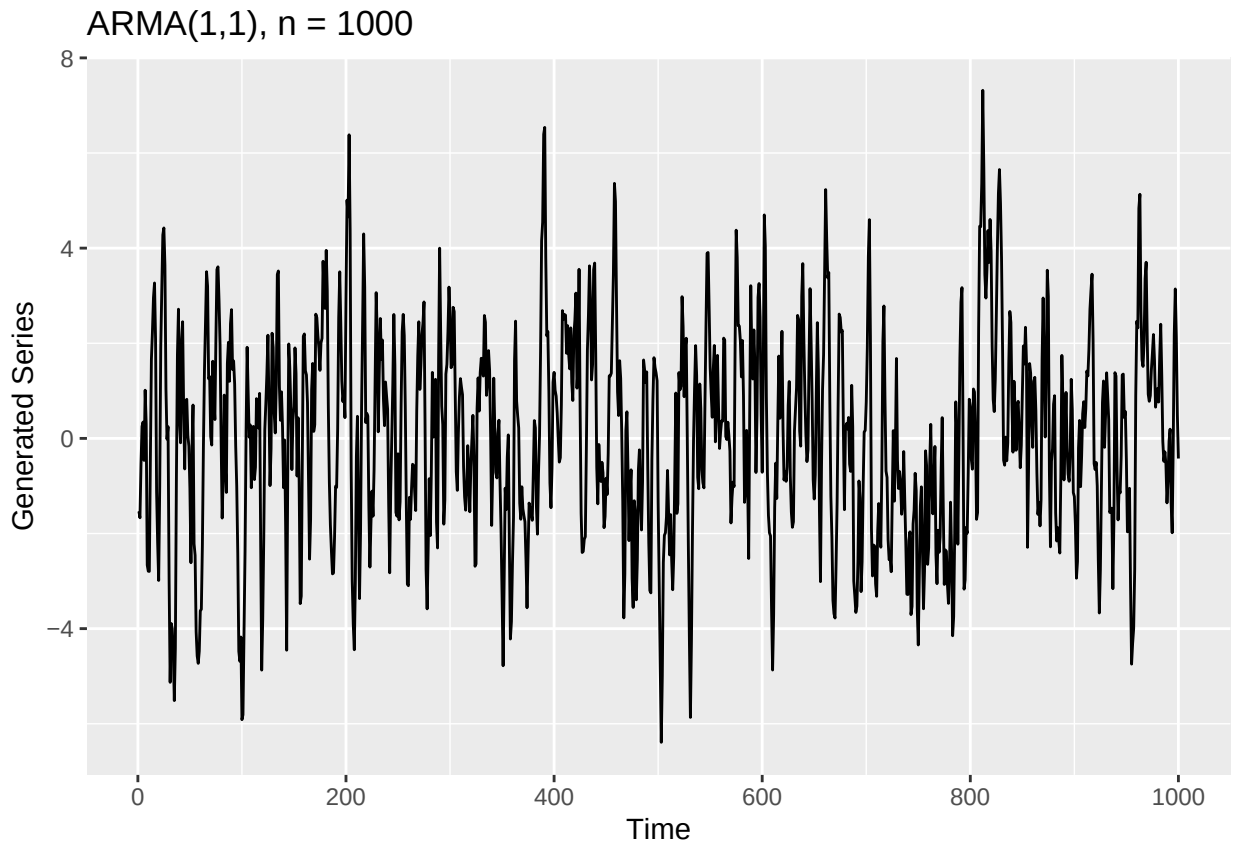
```r
ARMA10.b <- arima.sim(list(order=c(1,0,0),ar=c(0.6)),n=1000)
ARMA01.b <- arima.sim(list(order=c(0,0,1),ma=c(0.9)),n=100)
ARMA11.b <- arima.sim(list(order=c(1,0,1),ar=c(0.6),ma=c(0.9)),n=1000)

autoplot(ARMA10.b)+
  ggtitle("ARMA(1,0), n = 1000")+
  ylab("Generated Series")
```




```
autoplot(ARMA01.b)+
  ggtitle("ARMA(0,1), n = 1000")+
  ylab("Generated Series")
```

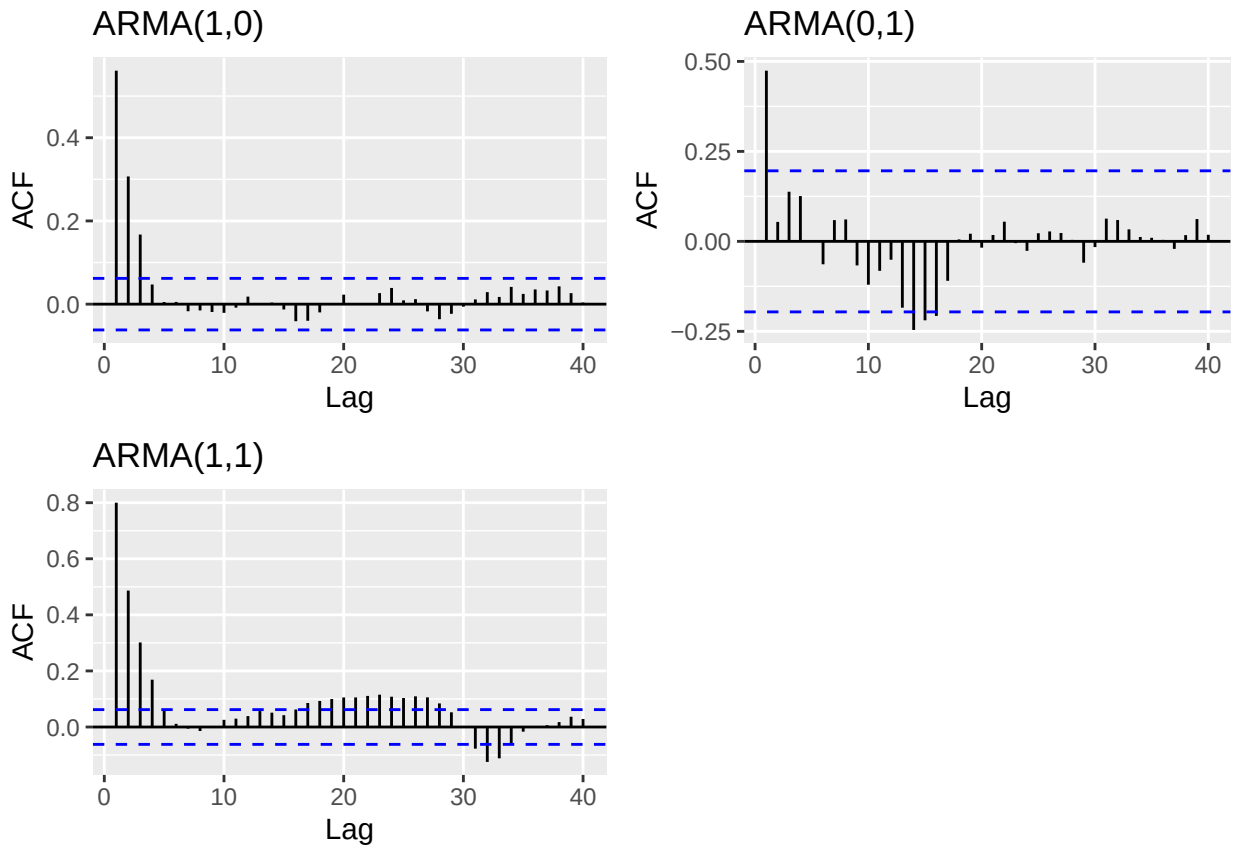



```
autoplot(ARMA11.b)+
  ggtitle("ARMA(1,1), n = 1000")+
  ylab("Generated Series")
```



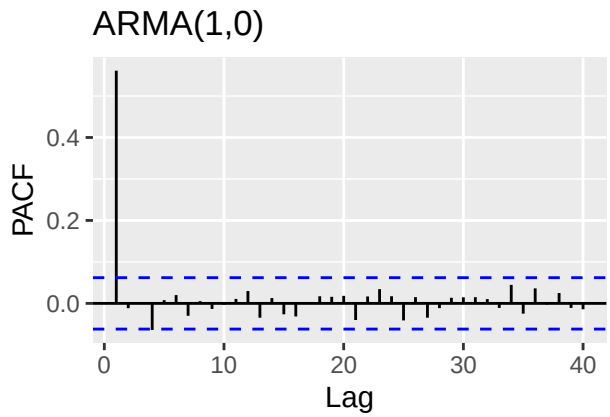

```r
#b
plot_grid(
  autoplot(Acf(ARMA10.b, lag.max=40, plot = FALSE),main = "ARMA(1,0)"),
  autoplot(Acf(ARMA01.b, lag.max=40, plot = FALSE), main = "ARMA(0,1)"),
  autoplot(Acf(ARMA11.b, lag.max=40, plot = FALSE),main = "ARMA(1,1)")
)
```

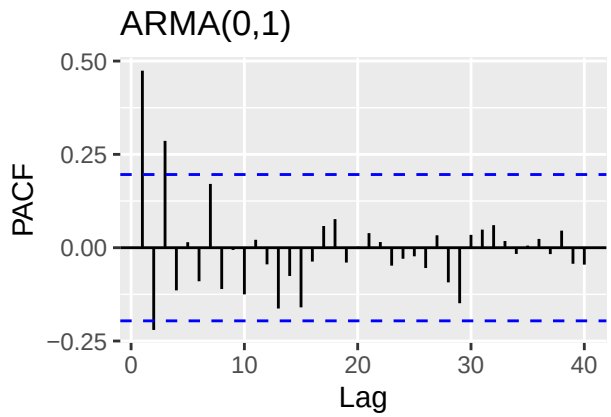
```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown parameters: `main`
## Ignoring unknown parameters: `main`
## Ignoring unknown parameters: `main`
```

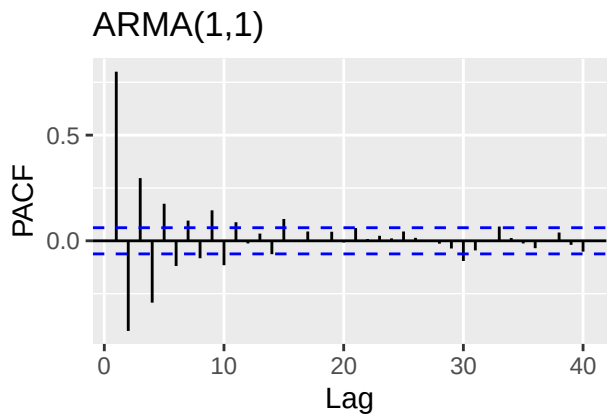



```r
#c
plot_grid(
  autoplot(Pacf(ARMA10.b, lag.max=40, plot = FALSE),main = "ARMA(1,0)"),
  autoplot(Pacf(ARMA01.b, lag.max=40, plot = FALSE), main = "ARMA(0,1)"),
  autoplot(Pacf(ARMA11.b, lag.max=40, plot = FALSE),main = "ARMA(1,1)")
)
```

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown parameters: `main`
## Ignoring unknown parameters: `main`
## Ignoring unknown parameters: `main`
```

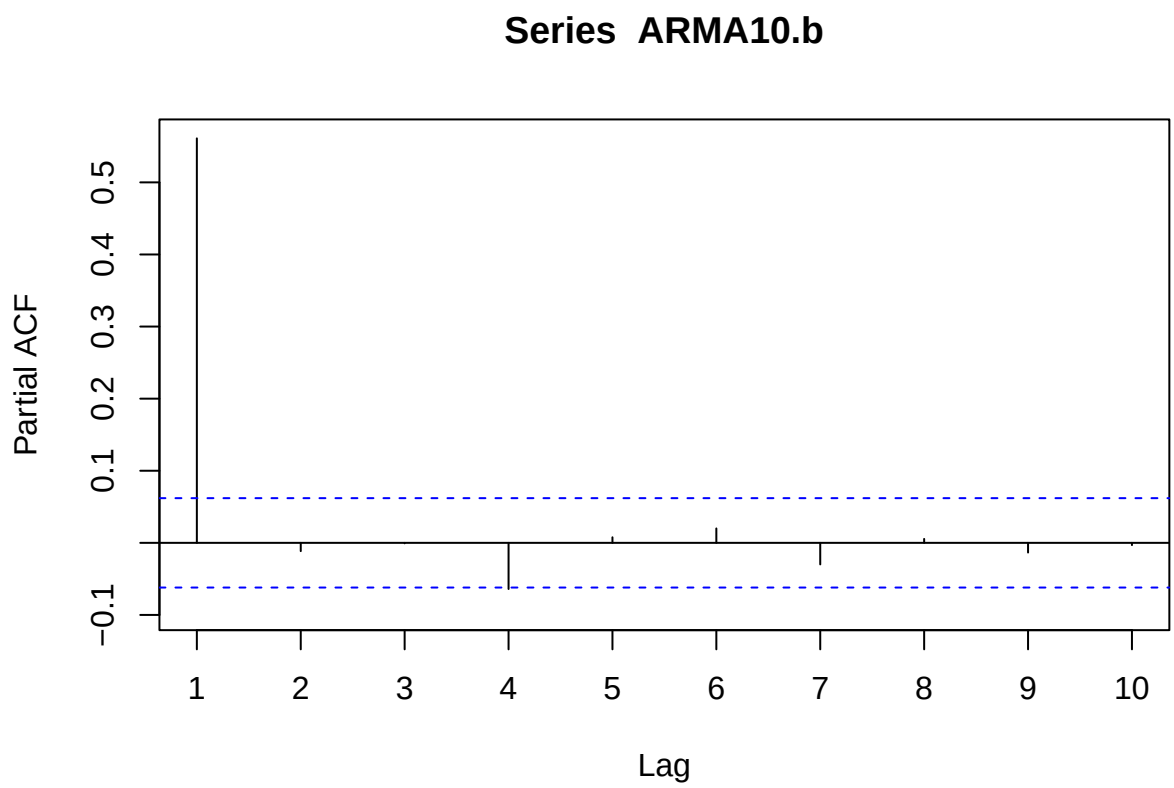






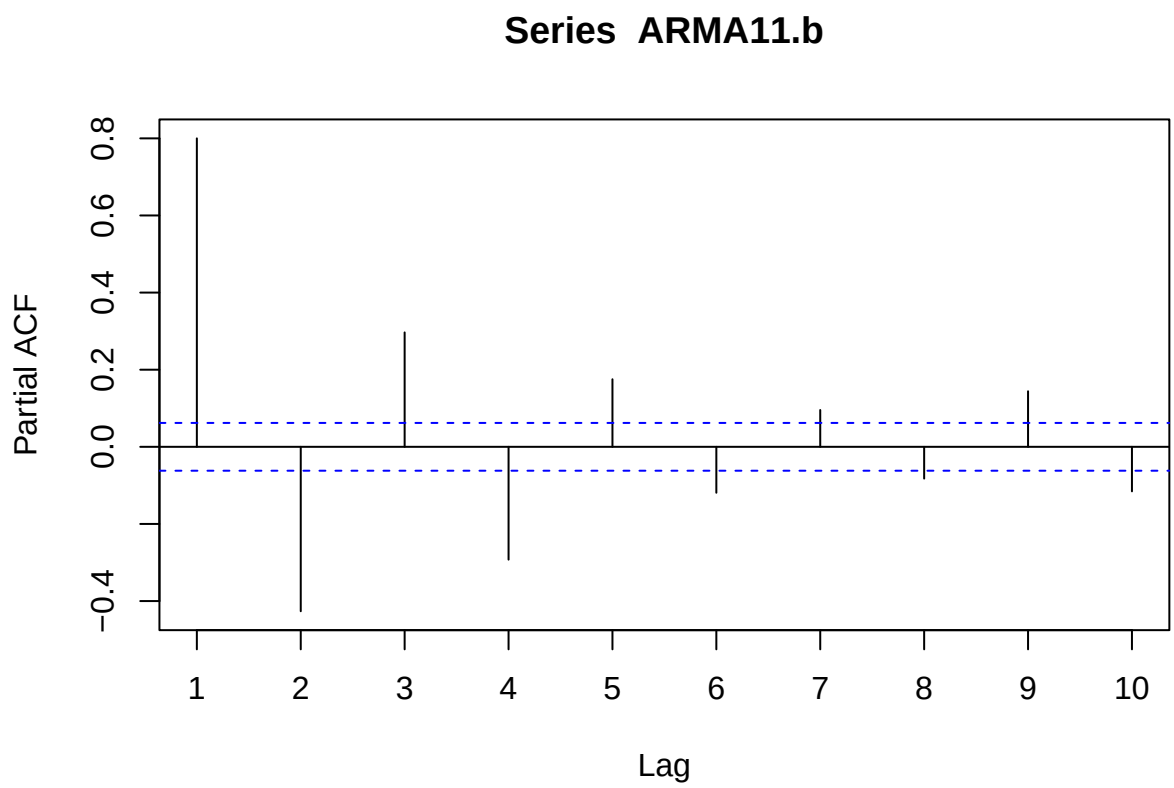
d) For the ARMA(1,0) model, the identification is obvious as AR(1) because of the cut off at lag 1 in the PACF and the slow decay in the ACF. The ARMA(0,1) ACF has one significant peak, and, not very distinct but present, slow decline in the PACF, so I would have identified it as MA(1). The ARMA(1,1) is very clearly ARMA because there is a clear slow decay for both the ACF and PACF. Overall a higher n yielded more easily identifiable models.

```
#e
print(Pacf(ARMA10.b, lag.max = 10))
```




```
##
## Partial autocorrelations of series 'ARMA10.b', by lag
##
##      1      2      3      4      5      6      7      8      9     10
##  0.561 -0.012 -0.001 -0.064  0.008  0.020 -0.030  0.005 -0.013 -0.003
```

```r
print(Pacf(ARMA11.b, lag.max = 10))
```




```
##
## Partial autocorrelations of series 'ARMA11.b', by lag
##
##      1      2      3      4      5      6      7      8      9     10
##  0.800 -0.426  0.297 -0.293  0.175 -0.119  0.096 -0.082  0.144 -0.115
```

e) Again, the PACF values at lag 1 for ARMA(1,0) and ARMA(1,1) are not equal.

Q3

Consider the ARIMA model $y_t = 0.7 * y_{t-1} - 0.25 * y_{t-12} + a_t - 0.1 * a_{t-1}$

(a) Identify the model using the notation ARIMA$(p, d, q)(P, D, Q)_s$, i.e., identify the integers p, d, q, P, D, Q, s (if possible) from the equation.

ARIMA(1,1,1)(1,1,0)

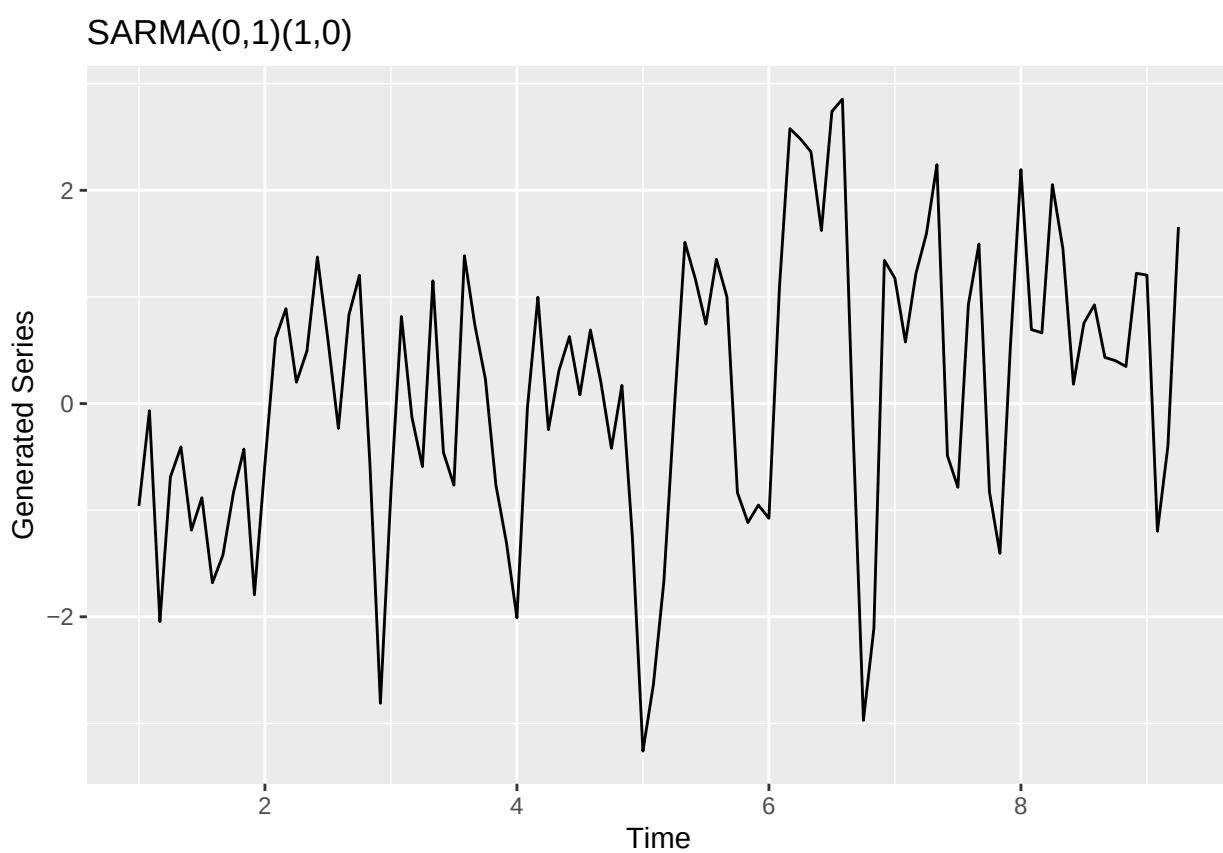
(b) Also from the equation what are the values of the parameters, i.e., model coefficients.

AR coefficient = 0.7, SAR coefficient = -0.25 MA coefficient = 0.1

Q4

Simulate a seasonal ARIMA$(0,1)\times(1,0)_{12}$ model with $\phi = 0.8$ and $\theta = 0.5$ using the `sim_sarima()` function from package `sarima`. The 12 after the bracket tells you that $s = 12$, i.e., the seasonal lag is 12, suggesting monthly data whose behavior is repeated every 12 months. You can generate as many observations as you like. Note the Integrated part was omitted. It means the series do not need differencing, therefore $d = D = 0$. Plot the generated series using `autoplot()`. Does it look seasonal?

```
sarima <- sim_sarima(n = 100, model = list(
    ma = 0.8,
    sar = 0.5,
    nseasons = 12
  )
)

#sim_sarima() gave me a numeric vector and autoplot requires a ts object
ts_sarima<-ts(sarima, frequency = 12)

autoplot(ts_sarima)+
  ggtitle("SARMA(0,1)(1,0)")+
  ylab("Generated Series")
```



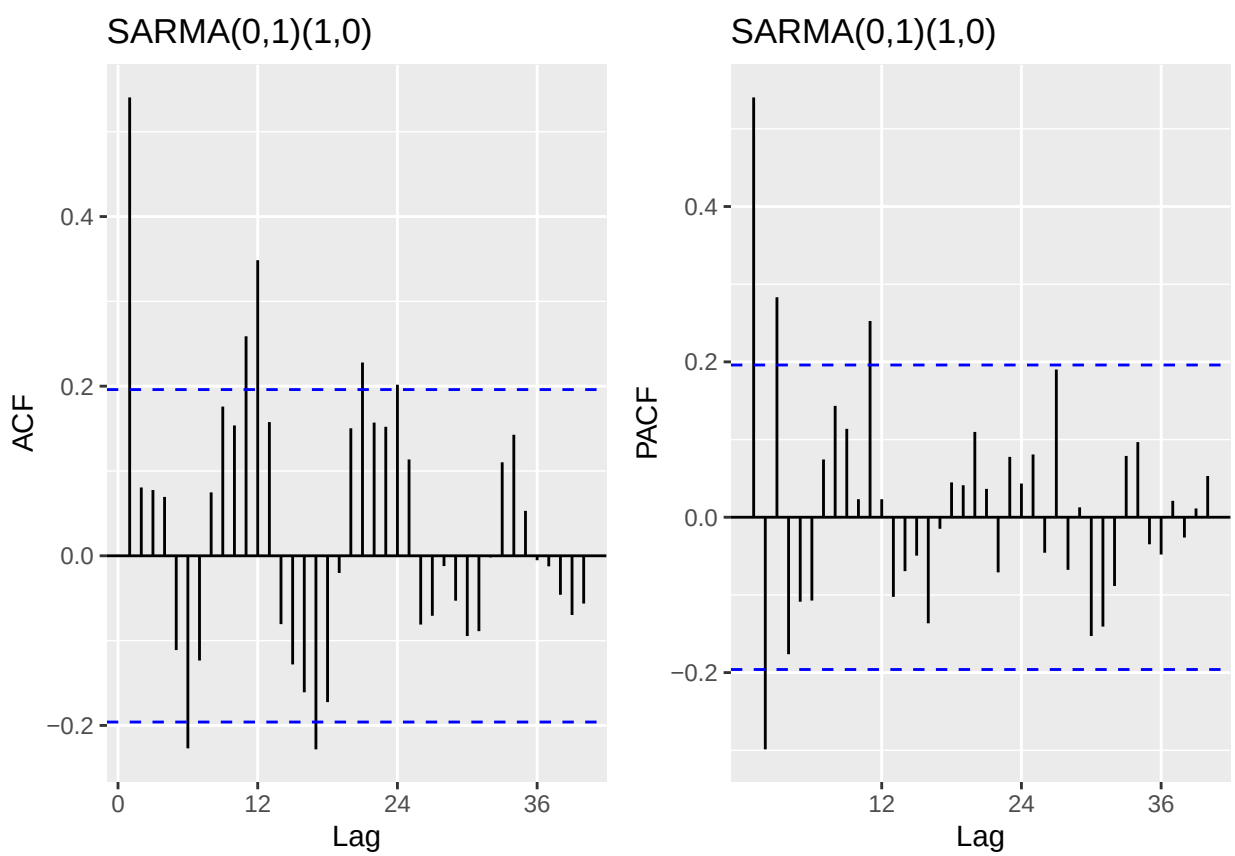
The resulting plot does not look distinctly seasonal. There are peaks and troughs but they vary quite a bit as far as magnitude, slope, and frequency.

Q5

Plot ACF and PACF of the simulated series in Q4. Comment if the plots are well representing the model you simulated, i.e., would you be able to identify the order of both non-seasonal and seasonal components from the plots? Explain.

```
plot_grid(
  autoplot(Acf(ts_sarima, lag.max=40, plot = FALSE),main = "SARMA(0,1)(1,0)"),
  autoplot(Pacf(ts_sarima, lag.max=40, plot = FALSE),main = "SARMA(0,1)(1,0)")
)
```

```
## Warning in ggplot2::geom_segment(lineend = "butt", ...): Ignoring unknown parameters: `main`
## Ignoring unknown parameters: `main`
```



I would not have been able to identify the order because for the seasonal component, a seasonal AR component would show positive spikes at each of the seasonal marks (12, 24, etc) and only one positive spike in the PACF at lag 12. While the ACF and PACF have positive spikes, they don't show the expected pattern. The non-seasonal component would have been difficult to identify because there is no clear cut off in the ACF or slow decay in the PACF as I would expect for the non-seasonal MA component.